

Contact

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(LinkedIn)

Top Skills

Large Language Models (LLM)
RAG

Large Language Model Operations
(LLMOps)

Certifications

AWS Academy Graduate - AWS
Academy Machine Learning
Foundations

Building LLM Applications With
Prompt Engineering

DATAAMP MATPLOTLIB
CERTIFICATION

Data Science & Analytics Virtual
Experience Program

Career Essentials in Data Analysis
by Microsoft and LinkedIn

Publications

GAN (Generative Adversarial
Network)-based Image Super-
resolution: A Technical Perspective

Rudra Patel

Research Assistant @ NASA-Collaborative Robotics & AI | ARCS
Lab, CSUN | Machine Learning, ROS2
United States

Summary

As a Research Assistant at NASA-Collaborative Robotics & AI | ARCS Lab, CSUN, I developed autonomous navigation systems for NASA rover prototypes using ROS2 and reinforcement learning. I also engineered a generative AI pipeline (GANs, PyTorch) to simulate high-fidelity Martian terrain for NASA missions, enhancing obstacle detection and rover decision-making in planetary exploration. Additionally, I created 3D mapping tools (Unity3D, Gazebo) to visualize rover navigation data, improving team collaboration and strategic planning for complex terrain challenges.

Previously, I worked as a Machine Learning Engineer at NXON AI PVT LTD, where I provided embedded product development and design services, including embedded software development, FPGA/ASIC design and verification, and hardware design. I hold a Bachelor of Technology in Information Technology from Charotar University of Science and Technology and am currently pursuing a Master of Science in Computer Science at California State University, Northridge. I am passionate about applying my skills in machine learning, robotics, and data science to solve real-world problems and contribute to the advancement of technology.

Experience

NASA-Collaborative Robotics & AI | ARCS Lab, CSUN
Research Assistant
February 2025 - Present (1 year 2 months)
United States

Developed autonomous navigation systems for NASA rover prototypes using ROS2 and reinforcement learning, enabling efficient pathfinding in Martian-like terrains during NASA-funded field tests.

- Engineered a generative AI pipeline (GANs, PyTorch) to simulate high-fidelity Martian terrain for NASA

missions, enhancing obstacle detection and rover decision-making in planetary exploration.

- Created 3D mapping tools (Unity3D, Gazebo) to visualize rover navigation data, improving team collaboration and strategic planning for complex terrain challenges.
- Collaborated with NASA JPL engineers to deploy SLAM algorithms on rover hardware, advancing real-time environment analysis capabilities for space exploration.
- Designed a computer vision module (OpenCV, YOLOv8) to identify mission-critical landmarks, supporting NASA objectives in resource detection and terrain analysis.

ISRO - Indian Space Research Organization

Data Science Intern

January 2024 - May 2024 (5 months)

Ahmedabad, Gujarat, India

NXON AI PVT LTD.

Machine Learning Engineer

May 2021 - November 2023 (2 years 7 months)

NXON Technologies is a group of Solution Team that solves engineering challenges by providing Embedded product development and design services including embedded software development, FPGA / ASIC design & verification and hardware design.

Medkart Pharmacy

Data Analyst

May 2023 - August 2023 (4 months)

Ahmedabad, Gujarat, India

Education

California State University, Northridge

Master of Science - MS, Computer Science · (August 2024 - September 2026)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Technology - BTech, Information Technology · (2020 - 2024)

Lokmanya vidyalaya higher secondary school science stream

Higher secondary school, Science A group · (January 2018)

